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Statistical Modelling and Inference

Week 1, Lecture 2

Outline:

- i. Confidence intervals
- ii. Testing of hypotheses
- iii. Power calculations

Question Left in Previous lecture

Let $Y_1, Y_2, \dots, Y_n$ be a random sample with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2$. Show that

$$S'^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

is a biased estimator for σ^2 and that

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

is an unbiased estimator for σ^2 .

Inferential Methods for Parameters

- **Parameter:** Numeric Description of a **Population**
- **Statistic:** Numeric Description of a **Sample**
- **Statistical Inference:** Use of observed **statistics** from sample to make statements regarding population **parameters**.
 - **Estimation:** Predicting the unknown parameter based on sample data. Can be either a **point estimate** (single number) or **interval estimate** (a range of values)
 - **Testing of Hypothesis:** Using sample data to see whether we can rule out specific values of an unknown parameter with a certain level of confidence

Estimation of Confidence Interval for Population Mean

- Although the parameter is unknown, it's highly likely that our sample mean (**estimate**) will lie within 2 standard deviations (**standard errors**) of the population mean (**parameter**).
- **Margin of Error**: Measure of the upper bound in sampling error with a fixed level (we will typically use 95%) of confidence. That will correspond to 2 standard errors:

$$\text{Margin of Error (95\% Confidence)} : 2 \frac{\sigma}{\sqrt{n}}$$

Confidence Interval : estimate $\pm$ margin of error

Estimation of Confidence Interval for Population Mean

- Goal: Estimate a population mean based on sample mean
- Unknown: Parameter (μ)
- Known: Approximate Sampling Distribution of Statistic

$$\bar{Y} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

For a random variable that is normally distributed, the probability that it will fall within 2 standard deviations of mean is approximately 0.95

$$P\left(\mu - 2 \frac{\sigma}{\sqrt{n}} \leq \bar{Y} \leq \mu + 2 \frac{\sigma}{\sqrt{n}}\right) \approx 0.95$$

One-Sample Confidence Interval for *mean* μ

- SRS from a population with mean μ is obtained.
- Sample mean, sample standard deviation are obtained
- Degrees of freedom are $df = n-1$, and confidence level $(1-\alpha)$ are selected
- Level $(1-\alpha)$ confidence interval of form:

$$\bar{y} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$t_{\alpha/2}$ selected from t - table so that $P(-t_{\alpha/2} \leq t(n-1) \leq t_{\alpha/2}) = 1 - \alpha$

Procedure is theoretically derived based on normally distributed data, but has been found to work well regardless for large n

Confidence Interval for a Population Mean μ

- Confidence Coefficient ($1-\alpha$): Probability (based on repeated samples and construction of intervals) that a confidence interval will contain the true mean μ
- Common choices of $1-\alpha$ and resulting intervals:

$$90\% \text{ Confidence : } \bar{y} \pm 1.645 \frac{\sigma}{\sqrt{n}}$$

$$95\% \text{ Confidence : } \bar{y} \pm 1.960 \frac{\sigma}{\sqrt{n}}$$

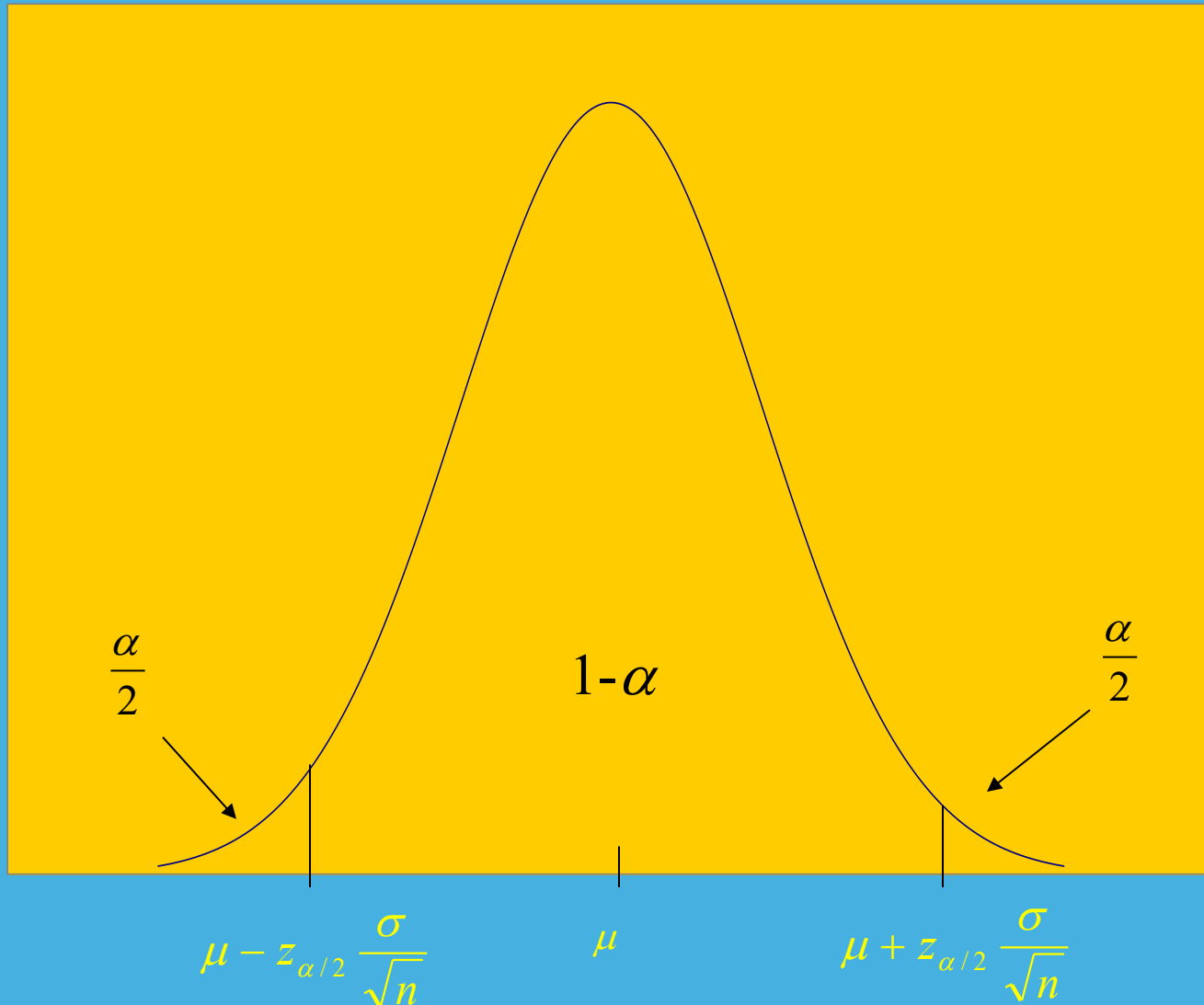
$$99\% \text{ Confidence : } \bar{y} \pm 2.576 \frac{\sigma}{\sqrt{n}}$$

$$(1-\alpha)100\% \text{ Confidence : } \bar{y} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$1-\alpha$	$\alpha/2$	$1-\alpha/2$	$z_{\alpha/2}$
0.90	0.050	0.950	1.645
0.95	0.025	0.975	1.960
0.99	0.005	0.995	2.576

Confidence Interval for a Population Mean μ

Normal Distribution



Example: 90% Confidence Interval for a Population Mean μ

The shopping times of $n = 64$ randomly selected customer at a supermarket were recorded. The mean shopping time of this sample is 33 minutes. Suppose that the variance of shopping time is 256 minutes, what is the 90% two-sided confidence interval for the true average shopping time per customer?

Let y is the shopping time in minutes.

$\bar{y} = 33$ mean shopping time

$\sigma^2 = 256$ variance of shopping time ($\sigma = 16$)

90% confidence interval

$$33 \pm 1.645 \frac{16}{\sqrt{64}}$$

$$33 \pm 3.29$$

$$(29.71, 36.29)$$

Example: 95% Confidence Interval for a Population Mean μ

Suppose market manager want to estimate the average weight of a specific variety of apple. He take a random sample of 36 apples and find the sample mean 150 grams with population standard deviation 12 grams. What is a 95% confidence interval for average apples weight?

$\bar{y} = 150$ given sample mean weight

$\sigma = 12$ population standard deviation

$$150 \pm 1.96 \frac{12}{\sqrt{36}}$$

$$150 \pm 3.92$$

$$(153.92, 146.08)$$

Question: 99% Confidence Interval for a Population Mean μ

Q1. A researcher needs to measures the average heights of men in a city. He select a sample of 150 randomly selected men and finds their average height is 175.5 cm with a standard deviation of 7.2 cm.

Find a 99% confidence interval of the average height of men.

Q2. A tech company wants to estimate the average battery life of its new smartphone model. They test a random sample of 100 phones and find a sample mean of 22 hours with a known population standard deviation 3.5 hours.

Find a 99% confidence interval to estimate the average battery life of its new smartphone model.

Hypothesis Testing

Statistical hypothesis

Statistical hypothesis testing of parameters are the fundamental methods used at the data analysis stage of a comparative experiment, in which we are interested, in comparing the population parameter to a specified value.

Definition:

A statistical hypothesis is a statement about the parameters of one or more populations.

If this information is consistent with the hypothesis, then we will conclude that the hypothesis is true; if this information is inconsistent with the hypothesis, we will conclude that the hypothesis is false.

Steps used in Hypothesis Testing

- **Null hypothesis (H_0):** Statement or theory being tested. Stated in terms of parameter(s) and contains an equality. Test is set up under the assumption of its truth.
- **Alternative Hypothesis (H_a):** Statement contradicting H_0 . Stated in terms of parameter(s) and contains an inequality. Will only be accepted if strong evidence refuse H_0 based on sample data. May be 1-sided or 2-sided, depending on theory being tested.
- **Test Statistic:** Quantity measuring difference between sample statistic (estimate) and parameter value under H_0
- **Rejection Region:** Values of test statistic for which we reject H_0 in favor of H_a
- **P-value:** Probability (assuming H_0 is true) that we would observe sample data (test statistic) this extreme or more extreme in favor of the alternative hypothesis (H_a)

Elements of a Hypothesis Testing

- **One-Tailed Test:** Test where the alternative hypothesis states specifically that the parameter is strictly above (below) the null value
- **Two-Tailed Test:** Test where the alternative hypothesis is that the parameter is not equal to null value (simultaneously tests “greater than” and “less than”)

Type I and type II Errors

- **Type I Error:** Test resulting in rejection of H_0 in favor of H_a when H_0 is in fact true
 - $P(\text{Type I error}) = \alpha$
- **Type II Error:** Test resulting in failure to reject H_0 in favor of H_a when in fact H_a is true (H_0 is false)
 - $P(\text{Type II error}) = \beta$

Test Statistic

- **Parameter:** Population mean (μ) under H_0 is μ_0
- **Statistic (Estimator):** Sample mean obtained from sample measurements is: $\bar{y}$
- **Standard Error of Estimator:** $S.E = \sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$
- **Sampling Distribution of Estimator:**
 - Normal if shape of distribution of individual measurements is normal
 - Approximately normal regardless of shape for large samples
- **Test Statistic:**

$$Z_{obs} = \frac{\bar{y} - \mu_0}{\sigma / \sqrt{n}}$$

Note: Normally σ is unknown and is replaced by s in large samples.

Decision Rules or Rejection Regions

- Once a significance (α) level has been chosen a decision rule can be stated, based on a critical value:
- 1-sided tests (Upper Tail): $H_0: \mu \leq \mu_0$ $H_a: \mu > \mu_0$
 - If test statistic $z_{\text{obs}} > z_{\alpha}$ Reject H_0 and conclude $\mu > \mu_0$
 - If $z_{\text{obs}} < z_{\alpha}$ Do not reject $H_0: \mu \leq \mu_0$
- 1-sided tests (Lower Tail): $H_0: \mu \geq \mu_0$ $H_a: \mu < \mu_0$
 - If test statistic $z_{\text{obs}} < -z_{\alpha}$ Reject H_0 and conclude $\mu < \mu_0$
 - If $z_{\text{obs}} > -z_{\alpha}$ Do not reject $H_0: \mu \geq \mu_0$
- 2-sided tests: $H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$
 - If test statistic $z_{\text{obs}} > z_{\alpha/2}$ Reject H_0 and conclude $\mu > \mu_0$
 - If test statistic $z_{\text{obs}} < -z_{\alpha/2}$ Reject H_0 and conclude $\mu < \mu_0$
 - If $-z_{\alpha/2} < z_{\text{obs}} < z_{\alpha/2}$ Do not reject $H_0: \mu = \mu_0$

Computing the P -Value

- 2-sided Tests: How likely is it to observe a sample mean as far of farther from the value of the parameter under the null hypothesis? ($H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$)

$$\text{Under } H_0 : \bar{Y} \sim N\left(\mu_0, \frac{\sigma}{\sqrt{n}}\right) \Rightarrow Z = \frac{\bar{Y} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1)$$

After obtaining the sample data, compute the mean and convert it to a z-score (z_{obs}) and find the area above $|z_{\text{obs}}|$ and below $-|z_{\text{obs}}|$ from the standard normal (z) table

For I-tail Test: Obtain the area above z_{obs} for upper tail tests ($H_a: \mu > \mu_0$) or below z_{obs} for lower tail tests ($H_a: \mu < \mu_0$)

Potential for Abuse of Tests

- Should choose a significance (α) level in advance and report test conclusion (significant/nonsignificant) as well as the P -value. Significance level of 0.05 is widely used in the academic literature
- Very large sample sizes can detect very small differences for a parameter value. A clinically meaningful effect should be determined, and confidence interval reported when possible
- A nonsignificant test result does not imply no effect (that H_0 is true).
- Many studies test many variables simultaneously. This can increase overall type I error rates

Inference for Population Mean

- Practical Problem: Sample mean has sampling distribution that is Normal with mean μ and standard deviation $\sigma / \sqrt{n}$ (when the data are normal, and approximately so for large samples). **σ is unknown.**
- Have an estimate of σ , s obtained from sample data. **Estimated standard error** of the sample mean is:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$
$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \sim t(n-1)$$

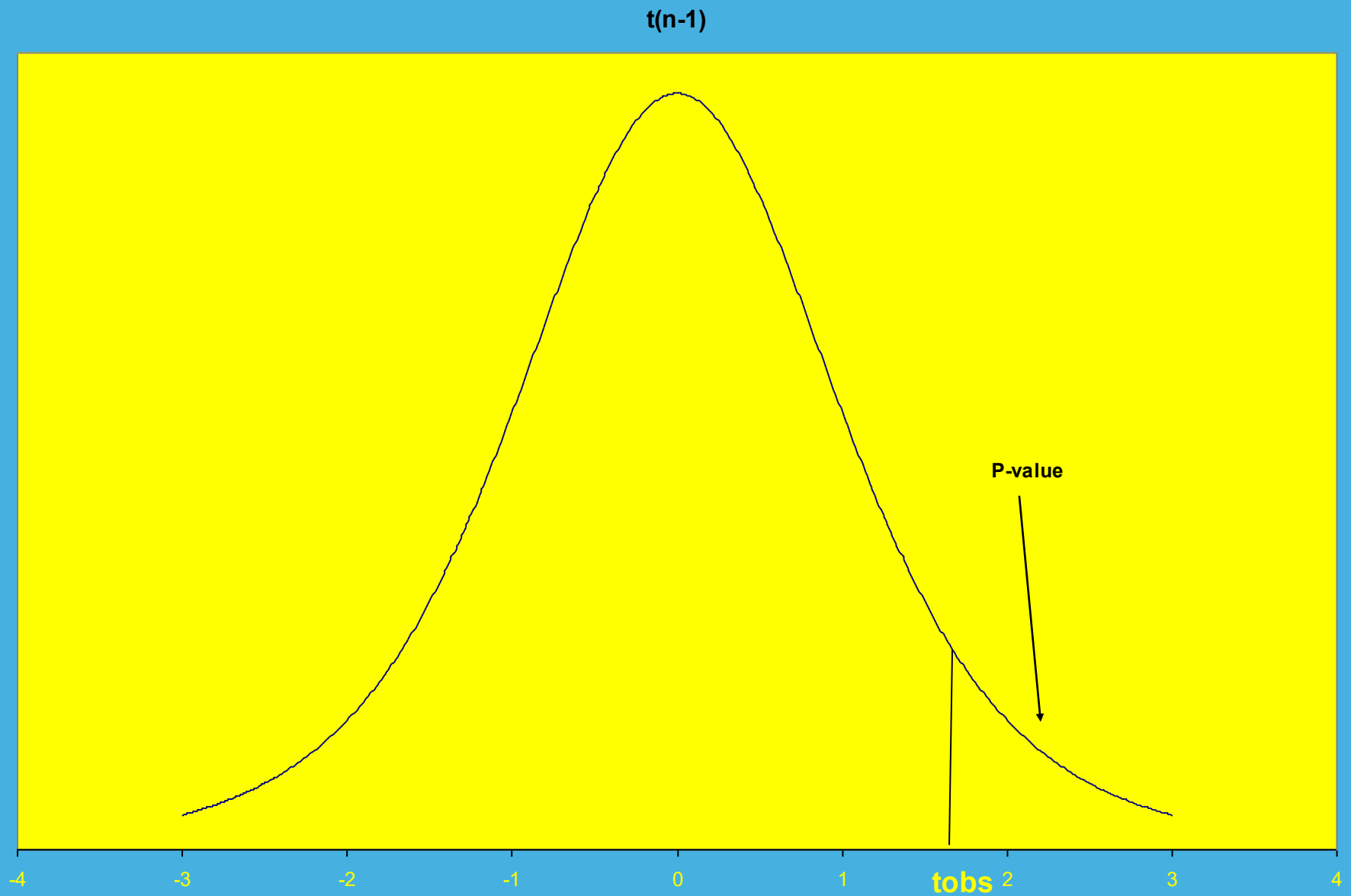
When the sample is SRS from $N(\mu, \sigma)$ then the t -statistic (same as z - with estimated standard deviation) is distributed t with $n-1$ degrees of freedom

Example of t -test (1-tailed or upper tailed)

- 1-sided Test: $H_0: \mu = \mu_0$ $H_a: \mu > \mu_0$
- Decision Rule (t_α such that $P(t(n-1) \geq t_\alpha) = \alpha$) :
 - Conclude $\mu > \mu_0$ if Test Statistic (t_{obs}) is greater than t_α
 - Do not conclude $\mu > \mu_0$ otherwise
- P -value: $P(t(n-1) \geq t_{obs})$
- Test Statistic:

$$t_{obs} = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

P -value (Upper Tail Test)



Example of t -test (1-tailed or lower tailed)

- 1-sided Test: $H_0: \mu = \mu_0$ $H_a: \mu < \mu_0$
- Decision Rule (t_α obtained such that $P(t(n-1) \geq t_\alpha) = \alpha$) :
 - Conclude $\mu < \mu_0$ if Test Statistic (t_{obs}) is less than $-t_\alpha$
 - Do not conclude $\mu < \mu_0$ otherwise
- P -value: $P(t(n-1) \leq t_{obs})$
- Test Statistic:

$$t_{obs} = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

Example of t -test (1-tailed or lower tailed)

- Testing whether population mean time to read list of colors is higher when color is written in different color ($\mu > 0$).
- Data: y_i : difference score for subject i (Different-Black)
- Null hypothesis (H_0): No difference effect ($H_0: \mu \leq 0$)
- Alternative hypothesis (H_a): difference effect ($H_a: \mu > 0$)
- $n = 70$ subjects in experiment, reasonably large sample

Sample Data : $\bar{y} = 2.39$ $s = 7.81$ $n = 70$

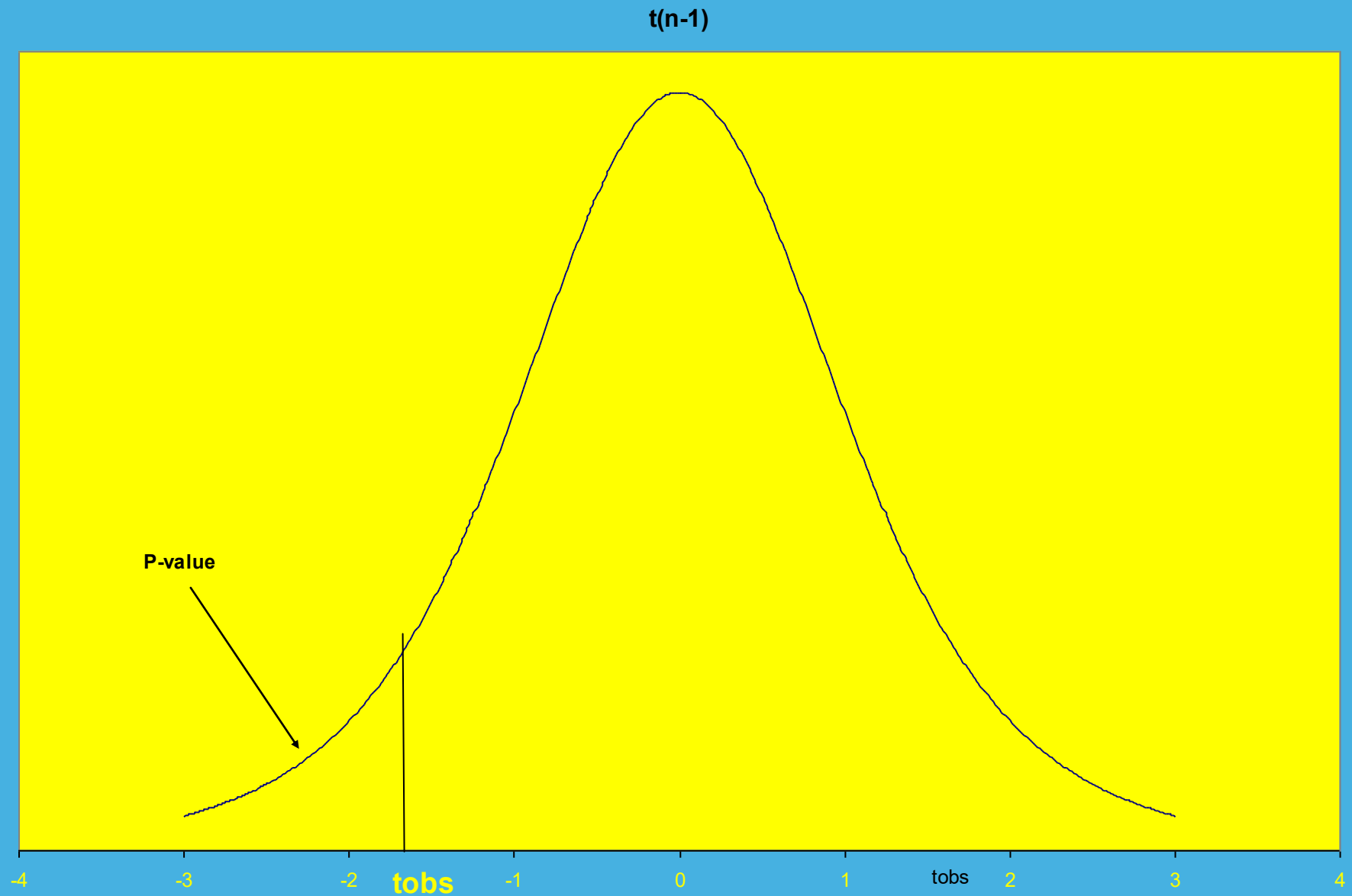
$$\text{Test Statistic (Based on } s = 7.81\text{)} : z_{obs} = \frac{2.39 - 0}{\left(\frac{7.81}{\sqrt{70}} \right)} = \frac{2.39}{0.93} = 2.57$$

Rejection Region ($\alpha = 0.05$): Reject H_0 if $z_{obs} \geq z_{.05} = 1.645$

P -value (Based on $s = 7.81$): $P(Z \geq 2.57) = 1 - .9949 = .0051$

Conclude there is evidence of difference effect ($\mu > 0$)

P -value (Lower Tail Test)



Example: t -test (2-sided or 2-tailed test)

- 2-sided Test: $H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$
- Decision Rule ($t_{\alpha/2}$ such that $P(t(n-1) \geq t_{\alpha/2}) = \alpha/2$) :
 - Conclude $\mu > \mu_0$ if Test Statistic (t_{obs}) is greater than $t_{\alpha/2}$
 - Conclude $\mu < \mu_0$ if Test Statistic (t_{obs}) is less than $-t_{\alpha/2}$
 - Do not conclude if $\mu \neq \mu_0$ otherwise
- P -value: $2P(t(n-1) \geq |t_{obs}|)$
- Test Statistic:

$$t_{obs} = \frac{\bar{y} - \mu_0}{s / \sqrt{n}}$$

Example: Mean Flight Time Beijing to Paris

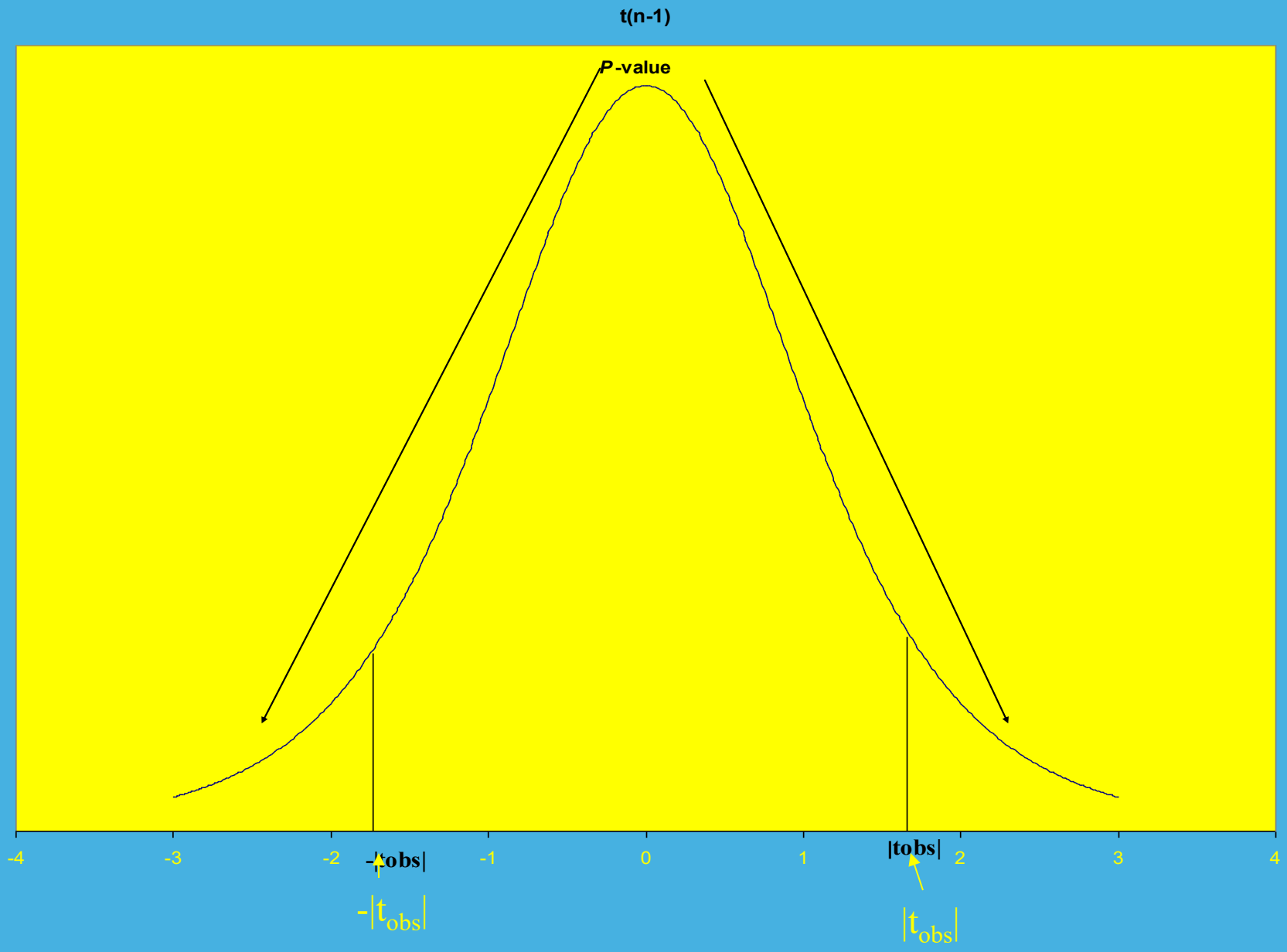
- Scheduled flight time: 580 minutes
- Sample: $n=31$ (selecting flights randomly from all flights)
- Test whether population mean flight time differs from scheduled time
- $H_0: \mu = 580$ $H_a: \mu \neq 580$
- Critical value (2-sided test, $\alpha = 0.05$, $n-1=30$ df): $t_{.025}=2.042$
- Sample data, Test Statistic, P -value:

$$\bar{y} = 574.1 \quad s = 19.7 \quad n = 31$$

$$t_{obs} = \frac{574.1 - 580}{19.7 / \sqrt{31}} = \frac{-5.9}{3.54} = -1.67$$

$$P\text{-value} = 2P(t(30) \geq |-1.67|) > 2P(t(30) \geq 1.697) = 2(.05) = .10$$

P-value (2-tailed test)



Two-sided Test

- Testing whether population mean time to read list of colors is effected (higher or lower) when color is written in different color
- Data: X_i : difference score for subject i (Different-Black)
- Null hypothesis (H_0): No difference effect ($H_0: \mu = 0$)
- Alternative hypothesis (H_a): difference effect (+ or -) ($H_a: \mu \neq 0$)

Sample Data : $\bar{x} = 2.39$ $s = 7.81$ $n = 70$

Test Statistic (Based on $s = 7.81$): $z_{obs} = \frac{2.39 - 0}{\left(\frac{7.81}{\sqrt{70}} \right)} = \frac{2.39}{0.93} = 2.57$

Rejection Region ($\alpha = 0.05$): $|z_{obs}| \geq z_{.05/2} = 1.96$

P -value (Based on $s = 7.81$): $2P(Z \geq |2.57|) = 2(1 - .9949) = .0102$

Again, there is evidence of difference effect ($\mu > 0$)

Power of a Test

- **Power** - Probability a test rejects H_0 (depends on μ)
 - H_0 True: Power = P(Type I error) = α
 - H_0 False: Power = 1-P(Type II error) = $1-\beta$
- **Example (Using context of in-difference data):**
 - $H_0: \mu = 0$
 - $H_A: \mu > 0$
 - $\sigma^2=64$ $n=16$
 - Decision Rule: Reject H_0 (at $\alpha=0.05$ significance level) if:

$$z_{obs} = \frac{\bar{y} - 0}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\bar{y}}{\sqrt{4}} \geq 1.645 \Rightarrow \bar{y} \geq 3.29$$

Power of a Test

- Now suppose that $\mu = 3.0$ (H_A is true)
- Power now refers to the probability we (correctly) reject the null hypothesis. Note that the sampling distribution of the sample mean is approximately normal, with mean 3.0 and standard deviation (standard error) 2.0.
- Decision Rule (from last slide): Conclude population mean interference effect is positive (greater than 0) if the sample mean difference score is above **3.29**
- Power for this case can be computed as:

$$P(\bar{y} \geq 3.29) \quad \text{when} \quad \bar{y} \sim N(3.0, 2.0)$$

Power of a Test

=> H_0 False: Power = 1-P(Type II error) = $1-\beta$

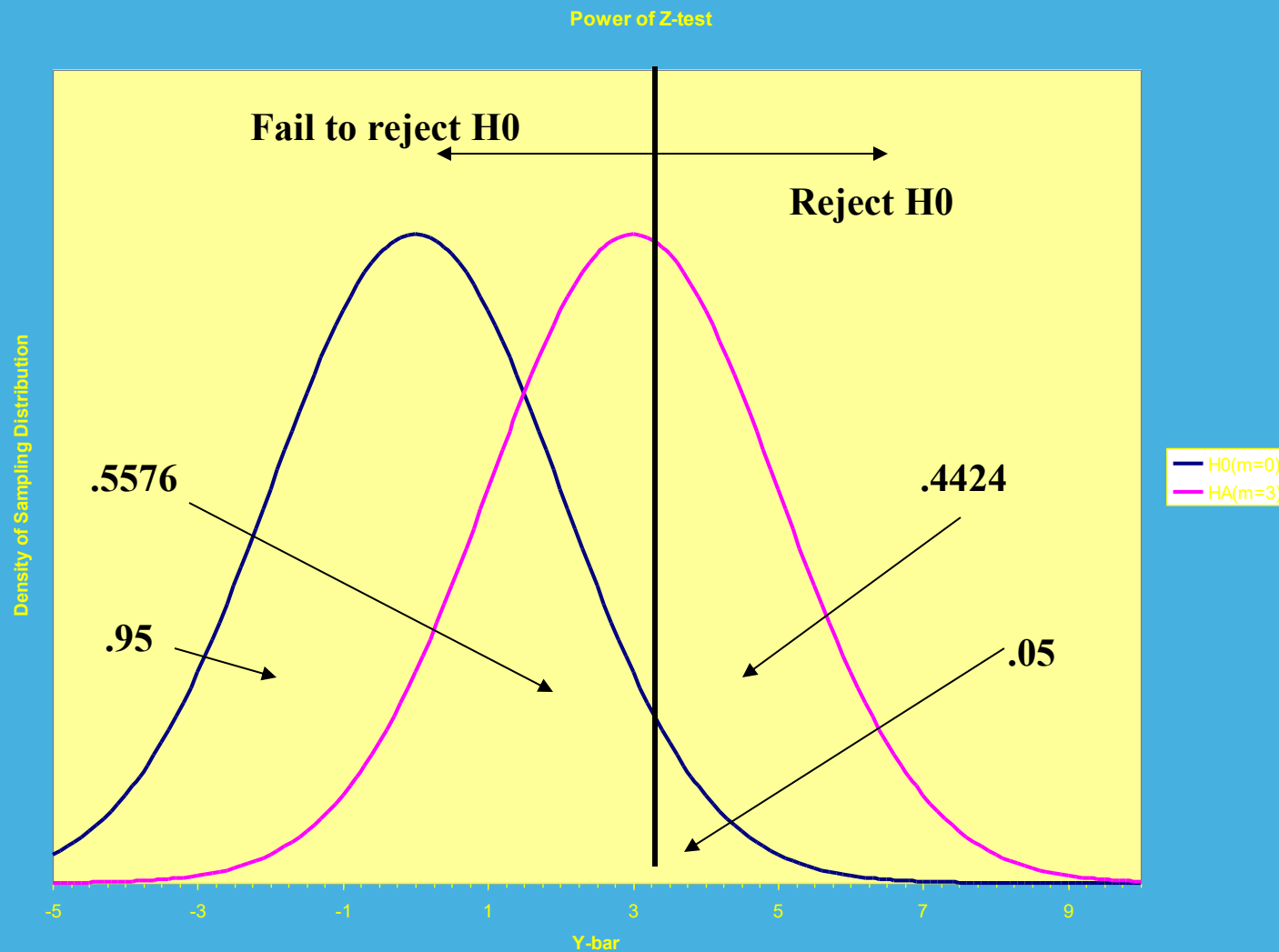
$$Power = P(\bar{y} \geq 3.29) = P\left(Z \geq \frac{3.29 - 3.0}{2.0} = 0.145\right) = 1 - .5576 = .4424$$

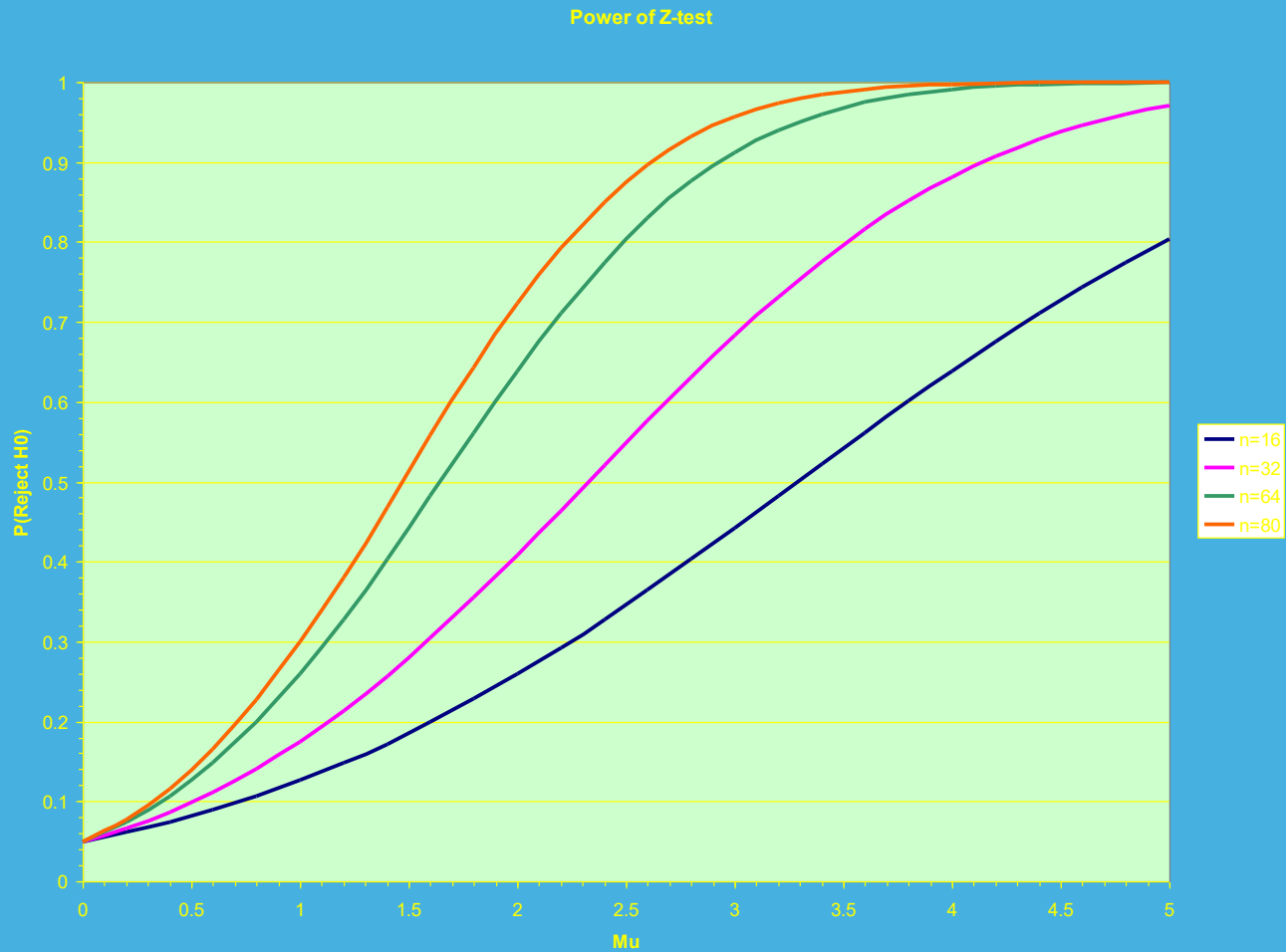
- All else being equal:
 - As sample size increases, power increases
 - As population variance decreases, power increases
 - As the true mean gets further from μ_0 , power increases

Power of a Test

Distribution (H_0)

Distribution (H_A)





- Power Curves for sample sizes of 16,32,64,80 and varying true values μ from 0 to 5 with $\sigma = 8$.
- For given μ , power increases with sample size
- For given sample size, power increases with μ

Example - Interference Data

- 2-Sided Test: $H_0:\mu = 0$ vs $H_a:\mu \neq 0$
- Set $\alpha = P(\text{Type I Error}) = 0.05$
- Choose important difference of $|\mu - \mu_0| = \Delta = 2.0$
- Choose Power = $P(\text{Reject } H_0 | \Delta = 2.0) = .90$
- Set $\beta = P(\text{Type II Error}) = 1 - \text{Power} = 1 - .90 = .10$
- From study, we know $\sigma \approx 8$

$$z_{\alpha/2} = z_{.05/2} = z_{.025} = 1.96 \quad z_{\beta} = z_{.10} = 1.282$$

$$n = \frac{\sigma^2}{\Delta^2} (z_{\alpha/2} + z_{\beta})^2 = \frac{(8)^2}{(2)^2} (1.96 + 1.282)^2 = 168.2 \approx 169$$

Would need 169 subjects to have a .90 probability of detecting effect

**Thank
You**

谢谢

